

Graphing Calculators are not permitted. A scantron answer sheet is provided. This is a formative assessment.

1. When $27y^3 - 125$ is factored completely, one of the factors is:

a) $3y + 5$ b) $y - 5$ c) $y + 5$ d) $9y^2 - 15y + 25$ e) $9y^2 + 15y + 25$

2. Simplify completely: $\frac{\frac{3y}{x+y} - \frac{1}{2}}{\frac{y}{x+y} - 1}$

a) $\frac{5}{2}$ b) $\frac{2y-x}{2x}$ c) $\frac{x-5y}{2x}$ d) $2y$ e) $2y + \frac{1}{2}$

3. Solve for x if $\frac{7}{x} - 4 = \frac{-4x}{x+1}$.

a) $-\frac{7}{3}$ b) $-\frac{7}{11}$ c) $-\frac{3}{7}$ d) $\frac{3}{7}$ e) $\frac{7}{3}$

4. Solve: $16^{2x+3} = 8^{x-1}$

a) -3 b) $-\frac{7}{3}$ c) $-\frac{1}{3}$ d) $\frac{1}{3}$ e) 3

5. Solve for x : $3^{3x} + 3^{3x+2} = 30$

a) -3 b) $-\frac{2}{3}$ c) $-\frac{1}{3}$ d) $\frac{1}{3}$ e) $\frac{2}{3}$

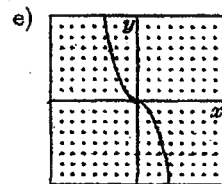
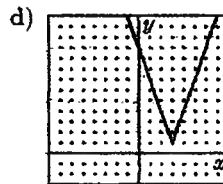
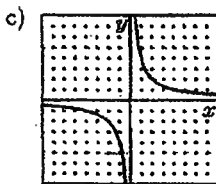
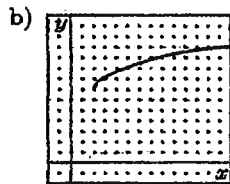
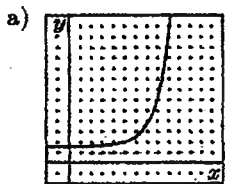
6. Consider the graph of $y = x^3$. It is slid 2 units to the right, then reflected in the x -axis, and finally slid 3 units up. What is the resulting equation?

a) $y = -(x+2)^3 + 3$ b) $y = (x-2)^3 + 3$ c) $y = -(x-2)^3 - 3$
d) $y = -(x-2)^3 + 3$ e) $y = (x-2)^{-3} + 3$

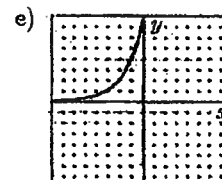
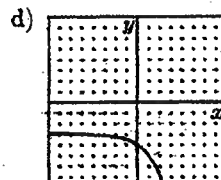
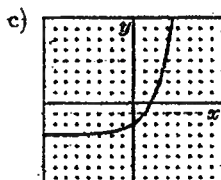
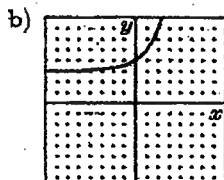
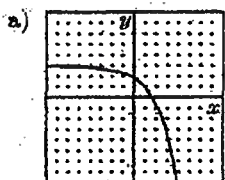
7. Consider the graph of $y = (x-1)(x+2)(x-3)$. If this graph is translated 3 units to the left, what is the sum of the x -intercepts?

a) -11 b) -7 c) -5 d) -1 e) 10

8. Which of the following is the graph of an exponential function?



9. Which one of the following sketches is a reasonable graph of $y = -2^x + 3$?



10. Which of the following equations would produce a graph with the general shape and position of the graph shown?

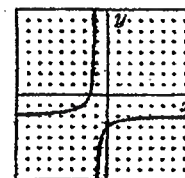
a) $y = \frac{2}{x+1} + 2$

b) $y = \frac{1}{x+1} - 2$

c) $y = \frac{2}{x-1} + 2$

d) $y = \frac{-1}{x+1} - 2$

e) $y = \frac{1}{x-1} - 2$



11. If $\sin x^\circ = \frac{1}{3}$, then what is the exact value of $\cos^2 x$?

a) $\frac{1}{9}$

b) $\frac{4}{9}$

c) $\frac{2}{3}$

d) $\frac{7}{9}$

e) $\frac{8}{9}$

12. Express $\frac{11\pi}{3}$ radians in degrees.

a) 145°

b) 300°

c) 330°

d) 630°

e) 660°

13. Convert 330° to radians. (Express your answer in terms of π .)

a) $\frac{\pi}{3}$

b) $\frac{5\pi}{3}$

c) $\frac{11\pi}{6}$

d) $\frac{10\pi}{3}$

e) $\frac{11\pi}{3}$

14. Convert 85° to radian measure accurate to the nearest hundredth

a) 0.12

b) 0.26

c) 1.48

d) 1.66

e) 4870.14

15. Given $\tan s = \sqrt{3}$ and $-\pi < s < 0$, find the value of s .

a) $-\frac{7\pi}{6}$

b) $-\frac{5\pi}{6}$

c) $-\frac{2\pi}{3}$

d) $-\frac{\pi}{3}$

e) $-\frac{\pi}{6}$

16. θ is an angle in standard position whose terminal arm is in quadrant IV and $\cos \theta = \frac{3}{\sqrt{13}}$. Find $\sin \theta$.

a) $-\frac{2}{\sqrt{13}}$

b) $-\frac{4}{\sqrt{13}}$

c) $\frac{2}{\sqrt{13}}$

d) $\frac{4}{\sqrt{13}}$

e) $\frac{\sqrt{13}}{2}$

17. Which of the following lines is an asymptote to the graph of $y = \tan(x - \frac{\pi}{4})$?

a) $x = 0$

b) $x = \frac{\pi}{2}$

c) $x = \frac{3\pi}{4}$

d) $x = \frac{5\pi}{4}$

e) $x = \pi$

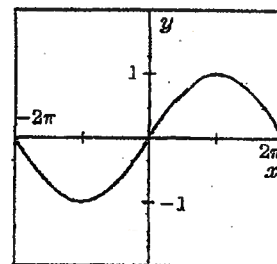
18. For $y = 23 \sin \frac{3\pi}{4}(x + 20) - 18$ state the (1) amplitude, (2) period, (3) vertical shift (positive or negative), and (4) phase displacement.

a) (1) 23, (2) $\frac{8}{3}$, (3) -18, (4) -20 b) (1) $\frac{8}{3}$, (2) 23, (3) -18, (4) -20 c) (1) $\frac{8}{3}$, (2) $\frac{3}{8}$, (3) 18, (4) -20

d) (1) 23, (2) $\frac{8}{3}$, (3) -20, (4) 18 e) (1) 23, (2) $\frac{8\pi}{3}$, (3) 18, (4) -20

19. Which of the following equations describes the graph shown?

- a) $y = 2 \sin \frac{\theta}{2}$ b) $y = \frac{1}{2} \sin \theta$ c) $y = \sin \theta$
 d) $y = 2 \sin 2\theta$ e) $y = \sin \frac{\theta}{2}$



20. What are the x -intercepts of the function $F(x) = -3 \cos(x + \pi) + 3$, if the domain is $0 \leq x < 2\pi$?

- a) π b) $0, \pi$ c) $\frac{\pi}{2}, \pi$ d) $\frac{\pi}{2}, \frac{3\pi}{2}$ e) $\frac{\pi}{3}, \frac{4\pi}{3}$

21. Simplify: $\sin x \csc x - \sin^2 x$

- a) 1 b) 0 c) $\cos^2 x$ d) $\sin^2 x$ e) $\sec x$

22. Simplify: $\frac{\csc A}{\cot A + \tan A}$

- a) $\cos A$ b) $\frac{1}{\sin^2 A \cos A}$ c) $\sin A$ d) $\frac{1}{\sin A}$ e) $\tan A$

23. Simplify: $\frac{2 \cos x}{\sin 2x}$

- a) 1 b) $\sec x$ c) $\cos x$ d) $\cos^2 x$ e) $\csc x$

24. Simplify the following to a single primary trigonometric function:

$$\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta}$$

- a) 1 b) 2 c) $2 \sec^2 \theta$ d) $2 \sin^2 \theta$ e) $2 \csc^2 \theta$

25. Which of the following is equivalent to $\cos(\theta - 270^\circ)$?

- a) $\tan \theta$ b) $-\cos \theta$ c) $\cos \theta$ d) $-\sin \theta$ e) $\sin \theta$

26. Simplify: $\cos\left(\frac{3\pi}{2} + \theta\right)$

- a) $\tan \theta$ b) $-\sin \theta$ c) $\sin \theta$ d) $-\cos \theta$ e) $\cos \theta$

27. Express $\sin(2\pi) \cos \frac{\pi}{6} - \cos(2\pi) \sin \frac{\pi}{6}$ as a trigonometric function of a single angle and simplify.

- a) $-\frac{1}{2}$ b) $-\frac{\sqrt{3}}{2}$ c) $\frac{1}{2}$ d) $\frac{\sqrt{3}}{2}$ e) $\frac{2\sqrt{3}}{2}$

28. Which of the following is equivalent to $\cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4}$?

- a) $\cos \frac{\pi}{12}$ b) $\cos \frac{\pi}{6}$ c) $\cos \frac{7\pi}{6}$ d) $-\sin \frac{7\pi}{6}$ e) $\sin \frac{7\pi}{6}$

29. Given that $\sin \alpha = -\frac{8}{17}$ and $\cos \beta = -\frac{4}{5}$, α and β are in quadrant III, then $\sin(\alpha - \beta) =$ _____.

- a) $-\frac{28}{85}$ **b) $-\frac{13}{85}$** c) $\frac{13}{85}$ d) $\frac{28}{85}$ e) $\frac{77}{85}$

30. Solve the following equation for θ , $0 \leq \theta < 2\pi$.

$$2\sin^2\theta + \sin\theta = 1$$

- a) $0, \frac{\pi}{2}, \pi$ b) $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{2}$ c) $\frac{2\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}$ d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ e) $\frac{\pi}{6}, \frac{3\pi}{2}$

31. Solve $2\cos\theta + \tan\theta = \sec\theta$ where $0 \leq \theta < 2\pi$. Answer in terms of π .

- a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ b) $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ c) $\frac{\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ d) $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$ e) $\frac{7\pi}{6}, \frac{11\pi}{6}$

32. Solve $\cos^2 x - 3\sin^2 x = 0$ for x , where $0 \leq x < 2\pi$. Answer in terms π .

- a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ b) $\frac{\pi}{3}, \frac{2\pi}{3}$ c) $\frac{\pi}{3}$ only d) $\frac{4\pi}{3}$ only e) $\frac{\pi}{3}, \frac{4\pi}{3}$

33. Write $10^{-3} = 0.001$ in logarithmic form.

- a) $\log(-3) = 0.001$ b) $-3\log 10 = 0.001$ c) $\log_{10} 10 = -2$
d) $\log(0.001) = -3$ e) $\log_3 0.001 = 10$

34. Evaluate: $\log_5 25\sqrt{5}$

- a) $\frac{2}{3}$ b) 2 c) $\frac{5}{2}$ d) $\frac{7}{2}$ e) $\frac{9}{2}$

35. Evaluate: $\log_{\frac{1}{2}} 32$

- a) -10 b) -5 c) $\sqrt{5}$ d) 10 e) 5

36. Solve: $\log_{16} x = -\frac{3}{2}$

- a) -16 b) $-\frac{1}{64}$ c) $\frac{1}{64}$ d) $\frac{1}{16}$ e) 16

37. Solve for x : $\log_{5x} 1225 = 2$

- a) ± 7 b) 7 c) $7\sqrt{7}$ d) 49 e) 1225

38. State the domain of the function $\log_x y = 2$.

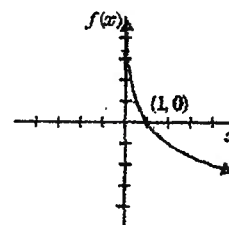
- a) $x > 0, x \neq 1$ b) $x > 0$ c) $x \geq 0$ d) $x \leq 0$ e) all real numbers

39. Which of the following is an asymptote for the graph of $y = 2^{x-1}$?

- a) $x = 1$ b) $x = 0$ c) $x = -1$ d) $y = 0$ e) $y = 1$

40. Which of the following equations could be represented by the graph shown?

- a) $y = 2^x$ b) $y = 2^{-x}$ c) $y = \log_2 x$
d) $y = -\log_2 x$ e) $y = \log_x 2$



41. Evaluate: $2\log_5 10 - \log_5 4$

- a) 1 b) 2 c) 14 d) 25 e) 5

42. Which of the following is equal to $\log_4 32 + \log_2 32$?
- a) $7\frac{1}{2}$ b) 2 c) 10 d) $12\frac{1}{2}$ e) 16
43. Which expression is equal to $\log \left(\frac{A}{B} \right)^2$?
- a) $\frac{1}{2} \log A - 2 \log B$ b) $\log A - 2 \log B$ c) $2 \log A - \log B$ d) $\log A + 2 \log B$ e) $2 \log A - 2 \log B$
44. Which of the following is equal to $\log(\frac{1}{2} \cdot \sin 2\theta)$?
- a) $\log(1 + \cos \theta) + \log(1 - \cos \theta)$ b) $-\log(\cos^2 \theta)$ c) $\log(\cos^2 \theta - 1)$
d) $\log(\sin \theta) + \log(\cos \theta)$ e) $\log(\sin^2 \theta)$
45. Solve $\log_b(2x + 1) - \log_b(x - 1) = \log_b 3$ for x .
- a) $\frac{4}{5}$ b) 2 c) $\frac{8}{5}$ d) 4 e) 5
46. Solve: $5^{2x+1} = 96$ (approximate the answer to 3 decimal places)
- a) 0.825 b) 0.918 c) 1.659 d) 2.836 e) 3.836
47. A radioactive material decays according to the formula $A = A_0 10^{-kt}$, where A is the final amount, A_0 is the initial amount and t is time in years. Find k , if 500 grams of this material decays to 450 grams in 10 years. (Accurate to 4 decimal places.)
- a) -16.9897 b) -4.4820 c) -0.9000 d) 0.0046 e) 1.1065
48. Sarah has \$12000 to invest and decides to invest it in a mutual fund that has averaged a compound rate of return of 15.2% for each of the last 6 years. How many years will it take for her money to grow to \$600000 if this fund continues to average 15.2% during the time invest? Round off your answer to the nearest year.
- a) 15 b) 21 c) 28 d) 43 e) 45
49. Find the quotient and remainder of $(x^3 + 8x^2 + 19x + 13) \div (x + 3)$.
- a) $(x^2 + 11x + 52)$ R13 b) $(x^2 + 5x + 4)$ R-1 c) $(x^2 + 11x + 52)$ R169
d) $(x^2 + 5x + 4)$ R1 e) $(x^2 + 11x + 52)$ R-1
50. What is the remainder when $3x^4 - x^2 + 3x + 1$ is divided by $x + 2$?
- a) 1 b) 24 c) 39 d) $\frac{295}{16}$ e) 753
51. Factor the polynomial $P(x) = x^3 - x^2 - 10x - 8$ completely.
- a) $(x + 2)(x + 1)(x + 4)$ b) $(x + 2)(x - 1)(x + 4)$ c) $(x - 2)(x + 1)(x + 4)$
d) $(x - 2)(x - 1)(x + 4)$ e) $(x - 4)(x + 1)(x + 2)$
52. If $(x + 1)$ is a factor of the polynomial $(x^3 + kx^2 + x + 6)$, then what is the value of k ?
- a) -8 b) -4 c) 2 d) 4 e) 8

53. Which of the following polynomial function written in factored form best represents the graph and table given in the figure?

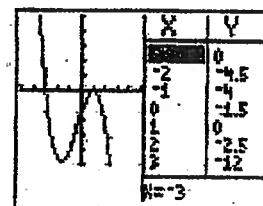
a) $P(x) = \frac{1}{2}(x-3)(x-1)^2$

b) $P(x) = \frac{1}{2}(x+3)(x-1)^2$

c) $P(x) = -\frac{1}{2}(x+3)(x-1)^2$

d) $P(x) = \frac{1}{2}(x-1)^3$

e) $P(x) = -\frac{1}{2}(x-1)^3$



54. Which of the following polynomial function written in factored form best represents the graph and table given in the figure?

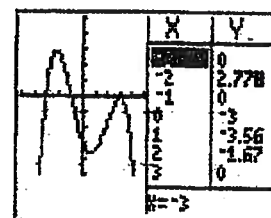
a) $P(x) = -\frac{1}{9}(x-3)^2(x+1)(x+3)$

b) $P(x) = -\frac{1}{9}(x+3)^2(x-1)(x-3)$

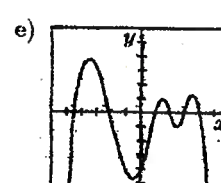
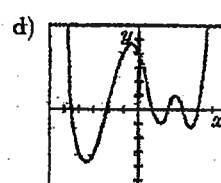
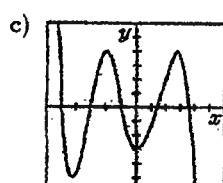
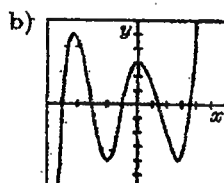
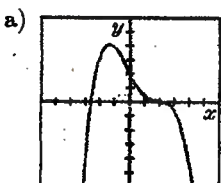
c) $P(x) = \frac{1}{9}(x-3)^2(x+1)(x+3)$

d) $P(x) = \frac{1}{9}(x+3)^2(x-1)(x-3)$

e) $P(x) = \frac{1}{9}(x+3)^2(x+1)(x-3)$



55. Which of the following graphs best illustrates the graph of a sixth degree polynomial function whose leading coefficient is negative?



56. Solve for x : $(x-2)(x+2)(x-6) \geq 0$

a) $-2 \leq x \leq 2$ or $x \geq 6$

b) $x \leq -6$ or $-2 \leq x \leq 2$

c) $-2 \leq x \leq 2$

d) $-6 \leq x \leq 2$

e) $x \leq -2$ or $2 \leq x \leq 6$

57. Solve for x : $x^3 - 7x + 6 < 0$

a) $-2 < x < -1$ or $x > 3$

b) $-3 < x < 1$ or $x > 2$

c) $-3 < x < -2$ or $x > 1$

d) $x < -3$ or $1 < x < 2$

e) $x > 3$

58. The figure shows the graph of f . Find $\lim_{x \rightarrow 10} f$.

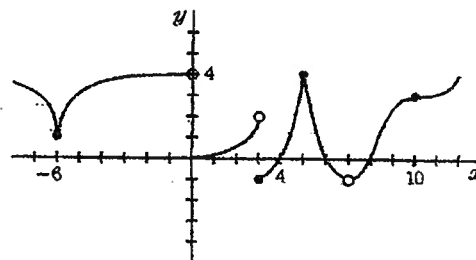
a) -1

b) 1

c) 2

d) 3

e) no limit



59. Refer to the previous figure. Find $\lim_{x \rightarrow 3} f$.

- a) -1 b) 1 c) 2 d) 3
e) no limit

60. Evaluate: $\lim_{x \rightarrow \infty} \frac{5x - 1}{2x}$

- a) 0 b) 1 c) $\frac{5}{2}$ d) 10 e) ∞

61. For the function $F(x) = \frac{3}{x}$, find the slope of the secant line when x changes from $x_1 = 6$ to $x_2 = 9$.

- a) $-\frac{1}{6}$ b) $-\frac{1}{18}$ c) $\frac{1}{18}$ d) $\frac{1}{6}$ e) $\frac{1}{2}$

62. Given the function $f(x) = x^2$, find $\frac{f(x+h) - f(x)}{h}$.

- a) $2x$ b) $2x + h$ c) $2x^2 + h$ d) $x^2 + h^2$ e) $3xh$